

"PAPER_{0:D3}" -f [int]# PAPER #88 ■ Neutrino SED: UQFF Emission Model

Author: Daniel T. Murphy

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Title: Neutrino Spectral Energy Distribution from Sgr A*: UQFF Multi-Flavor Emission Framework

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ($\tau = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$, $[\text{SCm}] = 0.99$)
Date: March 7, 2026 **Source Data:** MAIN1CoAnQi.cpp Batch 21 (NeutrinoSEModule), validate_phase3.py **Index Slot:** ■1.11 Black Hole Physics & Hawking Radiation, \$n = [int]# PAPER #88 ■ Neutrino SED: UQFF Emission Model

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Index Slot: ■1.11 Black Hole Physics & Hawking Radiation, PAPER088

Abstract

The neutrino spectral energy distribution (SED) from a black hole system ■ particularly Sagittarius A* ■ probes high-energy processes in the UQFF framework through three channels: Hawking neutrino emission ($T_{UQFF} = 0.99 \text{ TH}$), AGN corona pp/p? interactions (Ug4-mediated), and UQFF Toroidal Resonance Zone (TRZ) emission. Batch 21 of MAIN1CoAnQi.cpp implements the NeutrinoSEModule returning flux predictions at $E = 1 \text{ TeV} \text{---} 10 \text{ PeV}$ for comparison with IceCube-Gen2 sensitivity. The UQFF predicts a neutrino flux excess of 0.3% above the standard model corona prediction, driven by $f_{\text{TRZ}} = 0.01$.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \text{---} 10^4 \text{ day?}$ ■, $[\text{SSq}] = 0.57$) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Physical Origin of Neutrinos from BH Systems

Three UQFF neutrino production channels:

Channel	UQFF Term	Energy Range	Flux Contribution
Hawking emission	$T_{UQFF} = 0.99 \text{ TH}$	$E = \text{TH} \sim 10^4 \text{ K}$ (negligible)	Sub-dominant
Corona pp/p?	Ug4 AGN feedback	1 TeV ■ 10 PeV	Dominant
TRZ vacuum	$f_{\text{TRZ}} = 0.01$	0.1■100 PeV	1% enhancement

For stellar and SMBH systems, **Hawking neutrinos are negligible** ($\text{TH} \sim 10^4 \text{ K} \rightarrow E \sim \text{keV} \ll 1 \text{ eV}$). The interesting band is TeV■PeV cosmic neutrinos from AGN corona interactions.

2. The UQFF Neutrino SED

2.1 Standard Model Corona Prediction

For Sgr A* in active state with corona luminosity $L_{\text{corona}} = 10^{-4} L_{\text{Edd}}$:

$$\Phi_{\nu}^{\text{SM}}(E_{\nu}) = \Phi_0 \left(\frac{E_{\nu}}{\text{TeV}} \right)^{-\gamma} e^{-E_{\nu}/E_{\text{cut}}}$$

With $\gamma = 2.2$, $E_{\text{cut}} = 5 \text{ PeV}$.

2.2 UQFF Enhancement via TRZ

The Toroidal Resonance Zone factor $f_{\text{TRZ}} = 0.01$ enhances corona pair production:

$$\Phi_{\nu}^{\text{UQFF}}(E_{\nu}) = \Phi_{\nu}^{\text{SM}}(E_{\nu}) \times (1 + f_{\text{TRZ}}) = 1.01 \times \Phi_{\nu}^{\text{SM}}(E_{\nu})$$

2.3 Ug4-Mediated Enhancement

The Ug4 energy density ($3.352941 \times 10^{-12} \text{ J/m}^3$ baseline) additionally contributes to pion production:

$$\Delta \Phi_{\nu}^{\text{Ug4}}(E_{\nu}) = f_{\text{Ug4}} \cdot \Phi_{\nu}^{\text{SM}} \cdot \frac{U_{\text{g4}}}{U_{\text{g4,ref}}}$$

For current Sgr A* quiescent state ($A_{\text{AGN}} = 1$, $e^{-\tau}$ near-unity): $f_{\text{Ug4}} \ll f_{\text{TRZ}}$.

3. Multi-Flavor Ratio

Standard model predictions: $\nu_e : \nu_{\mu} : \nu_{\tau} = 1:2:0$ at production $\rightarrow 1:1:1$ after oscillation.

UQFF modification via 26D channel mixing (Batch 21):

$$(\nu_e : \nu_{\mu} : \nu_{\tau})_{\text{UQFF}} = (1 : 1 : 1) \times (1 + 0.001 \cdot f_{\text{TRZ}})$$

The 26D mixing is degenerate in flavor at the detector level **no measurable deviation from 1:1:1** at IceCube sensitivity. This prediction is consistent with IceCube observations showing approximate (but not exact) flavor democracy.

4. NeutrinoSEDMModule (Batch 21)

From MAIN1CoAnQi.cpp Batch 21:

```
// namespace SOURCE_BATCH21
// class NeutrinoSEDUQFFTerm : public PhysicsTerm
double NeutrinoSEDUQFFTerm::compute() const {
// Returns integrated flux: E^2 dF/dE at E = E_ref (1 TeV)
double SM_flux = phi0 * pow(E_ref/TeV, -gamma) * exp(-E_ref/E_cut);
double TRZ_factor = 1.0 + f_TRZ; // = 1.01
double Ug4_factor = 1.0 + f_Ug4 * Ug4_current / Ug4_ref;
return SM_flux * TRZ_factor * Ug4_factor;
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```

5. validate_phase3.py Validation

validate_phase3.py performs phase-3 cross-validation of Batch 21 outputs against uqff_validation_test.py:

- Neutrino SED test: UQFF flux = 2s deviation from IceCube diffuse background ? PASS
- Multi-flavor test: (1:1:1) ratio ? PASS
- TRZ enhancement: ? = 1.0% above SM ? PASS
- Ug4 quiescent: negligible Ug4 contribution at A_AGN = 1 ? PASS

6. IceCube-Gen2 Detectability

Current IceCube sensitivity: $F_{3s} \sim 10^{-10} \text{ TeV cm}^2 \text{ s}^{-1} \text{ sr}^{-1}$ for Sgr A* point source.

UQFF prediction: ? = 0.3% excess above SM corona prediction.

Not detectable by IceCube-Gen2 (~10-year run). However, an exaggerated AGN-active Sgr A* state ($A_{\text{AGN}} \gg 10$) would push Ug4 enhancement to ~5% potentially within 3s of Gen2.

Summary

Prediction	UQFF Value	Standard Model	UQFF Signature
TRZ enhancement	+1.0%	0%	UQFF-specific
Flavor ratio	1:1:1 (1.001)	1:1:1	Unmeasurable
Quiescent Ug4	negligible	None	Consistent
AGN-active Ug4	+0.3-5%	0%	Conditional
Phase 3 validation	4/4 PASS	N/A	Verified

Source: MAIN1CoAnQi.cpp Batch 21 (NeutrinoSEModule) | validatephase3.py | fTRZ = 0.01 | IceCube constraints.Groups[1].Value "PAPER_{0:D3}" -f \$n

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For stellar and SMBH systems, **Hawking neutrinos are negligible** ($TH \sim 10^9 - 4 K$? $E \sim k_B TH \ll 1$ eV). The interesting band is TeV-PeV cosmic neutrinos from AGN corona interactions.

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With $\gamma = 2.2$, $E_{\text{cut}} = 5$ PeV.

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The Ug4 energy density (3.352941×10^{-12} J/m³ baseline) additionally contributes to pion production:

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For current Sgr A* quiescent state ($A_{AGN} = 1$, $e^{-\tau}$ near-unity): $f_{Ug4} \ll f_{TRZ}$.

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- Multi-flavor test: (1:1:1) ratio ? PASS
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- Ug4 quiescent: negligible Ug4 contribution at A_AGN = 1 ? PASS

6. IceCube-Gen2 Detectability

Current IceCube sensitivity: F_3s 10^10 TeV cm^2 s^-1 sr^-1 for Sgr A* point source.

UQFF prediction: ? = 0.3% excess above SM corona prediction.

Not detectable by IceCube-Gen2 (~10-year run). However, an exaggerated AGN-active Sgr A* state (A_AGN >> 10) would push Ug4 enhancement to ~5% potentially within 3s of Gen2.

Summary

Prediction	UQFF Value	Standard Model	UQFF Signature
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Phase 3 validation	4/4 PASS	N/A	Verified

Source: MAIN1CoAnQi.cpp Batch 21 (NeutrinoSEDModule) | validatephase3.py | fTRZ = 0.01 | IceCube constraints .Groups[1].Value Neutrino SED: UQFF Emission Model

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Index Slot: 1.11 Black Hole Physics & Hawking Radiation, \$n = [int]\$ "PAPER{0:D3}" -f [int]\$ PAPER

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Three UQFF neutrino production channels:

Channel	UQFF Term	Energy Range	Flux Contribution
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Corona pp/p?	Ug4 AGN feedback	1 TeV ■ 10 PeV	Dominant
TRZ vacuum	$f_{\text{TRZ}} = 0.01$	0.1 ■ 100 PeV	1% enhancement

For stellar and SMBH systems, **Hawking neutrinos are negligible** ($T_{\text{H}} \sim 10^{-4} \text{ K} \Rightarrow E \sim k_B T_{\text{H}} \ll 1 \text{ eV}$). The interesting band is TeV ■ PeV cosmic neutrinos from AGN corona interactions.

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2.1 Standard Model Corona Prediction

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With $\gamma = 2.2$, $E_{\text{cut}} = 5$ PeV.

2.2 UQFF Enhancement via TRZ

The Toroidal Resonance Zone factor $f_{\text{TRZ}} = 0.01$ enhances corona pair production:

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The Ug4 energy density (3.352941×10^{10} J/m baseline) additionally contributes to pion production:

$$\Delta \Phi_{\nu}^{\text{Ug4}}(E_{\nu}) = f_{\text{Ug4}} \cdot \Phi_{\nu}^{\text{SM}} \cdot \frac{U_{\text{g4}}}{U_{\text{g4,ref}}}$$

For current Sgr A* quiescent state ($A_{\text{AGN}} = 1$, $e^{-\gamma t}$ near-unity): $f_{\text{Ug4}} \ll f_{\text{TRZ}}$.

3. Multi-Flavor Ratio

Standard model predictions: $\phi_e : \phi_{\mu} : \phi_{\tau} = 1:2:0$ at production $\rightarrow 1:1:1$ after oscillation.

UQFF modification via 26D channel mixing (Batch 21):

$$(\nu_e : \nu_{\mu} : \nu_{\tau})_{\text{UQFF}} = (1 : 1 : 1) \times (1 + 0.001 \cdot f_{\text{TRZ}})$$

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validate_phase3.py performs phase-3 cross-validation of Batch 21 outputs against uqff_validation_test.py:

Neutrino SED test: UQFF flux = 2s deviation from IceCube diffuse background ? **PASS**

Multi-flavor test: (1:1:1) ratio ? **PASS**

TRZ enhancement: $\gamma = 1.0\%$ above SM ? **PASS**

Ug4 quiescent: negligible Ug4 contribution at A_AGN = 1 ? **PASS**

6. IceCube-Gen2 Detectability

Current IceCube sensitivity: $F_{3s} \sim 10^{-10} \text{ TeV cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$ for Sgr A* point source.

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1. Physical Origin of Neutrinos from BH Systems

Three UQFF neutrino production channels:

Channel	UQFF Term	Energy Range	Flux Contribution
Hawking emission	$T_{\text{UQFF}} = 0.99 \text{ TH}$	$E \sim T_{\text{H}} \sim 10^{-4} \text{ K}$ (negligible)	Sub-dominant
Corona pp/p	Ug4 AGN feedback	1 TeV — 10 PeV	Dominant
TRZ vacuum	$f_{\text{TRZ}} = 0.01$	0.1 — 100 PeV	1% enhancement

For stellar and SMBH systems, **Hawking neutrinos are negligible** ($T_{\text{H}} \sim 10^{-4} \text{ K}$ $\Rightarrow E \sim k_B T_{\text{H}} \ll 1 \text{ eV}$). The interesting band is TeV–PeV cosmic neutrinos from AGN corona interactions.

2. The UQFF Neutrino SED

2.1 Standard Model Corona Prediction

For Sgr A* in active state with corona luminosity $L_{\text{corona}} = 10^{-4} L_{\text{Edd}}$:

$$\Phi_{\nu}^{\text{SM}}(E_{\nu}) = \Phi_0 \left(\frac{E_{\nu}}{\text{TeV}} \right)^{-\gamma} e^{-E_{\nu}/E_{\text{cut}}}$$

With $\gamma = 2.2$, $E_{\text{cut}} = 5 \text{ PeV}$.

2.2 UQFF Enhancement via TRZ

The Toroidal Resonance Zone factor $f_{\text{TRZ}} = 0.01$ enhances corona pair production:

$$\Phi_{\nu}^{\text{UQFF}}(E_{\nu}) = \Phi_{\nu}^{\text{SM}}(E_{\nu}) \times (1 + f_{\text{TRZ}}) = 1.01 \times \Phi_{\nu}^{\text{SM}}(E_{\nu})$$

2.3 Ug4-Mediated Enhancement

The Ug4 energy density ($3.352941 \times 10^{10} \text{ J/m}^3$ baseline) additionally contributes to pion production:

$$\Delta \Phi_{\nu}^{\text{Ug4}}(E_{\nu}) = f_{\text{Ug4}} \cdot \Phi_{\nu}^{\text{SM}} \cdot \frac{U_{\text{g4}}}{U_{\text{g4, ref}}}$$

For current Sgr A* quiescent state ($A_{\text{AGN}} = 1$, $e^{-\tau t}$ near-unity): $f_{\text{Ug4}} \ll f_{\text{TRZ}}$.

3. Multi-Flavor Ratio

Standard model predictions: $\nu_e : \nu_{\mu} : \nu_{\tau} = 1:2:0$ at production $\rightarrow 1:1:1$ after oscillation.

UQFF modification via 26D channel mixing (Batch 21):

$$(\nu_e : \nu_{\mu} : \nu_{\tau})_{\text{UQFF}} = (1 : 1 : 1) \times (1 + 0.001 \cdot f_{\text{TRZ}})$$

The 26D mixing is degenerate in flavor at the detector level **no measurable deviation from 1:1:1** at IceCube sensitivity. This prediction is consistent with IceCube observations showing approximate (but not exact) flavor democracy.

4. NeutrinoSEDMODULE (Batch 21)

From MAIN1CoAnQi.cpp Batch 21:

```
// namespace SOURCE_BATCH21
// class NeutrinoSEDUQFFTerm : public PhysicsTerm
double NeutrinoSEDUQFFTerm::compute() const {
// Returns integrated flux: E^2 dF/dE at E = E_ref (1 TeV)
double SM_flux = phi0 * pow(E_ref/TeV, -gamma) * exp(-E_ref/E_cut);
double TRZ_factor = 1.0 + f_TRZ; // = 1.01
double Ug4_factor = 1.0 + f_Ug4 * Ug4_current / Ug4_ref;
return SM_flux * TRZ_factor * Ug4_factor;
}
```

5. validate_phase3.py Validation

validate_phase3.py performs phase-3 cross-validation of Batch 21 outputs against uqff_validation_test.py:

Neutrino SED test: UQFF flux = 2s deviation from IceCube diffuse background ? **PASS**

Multi-flavor test: (1:1:1) ratio ? **PASS**

TRZ enhancement: $\tau = 1.0\%$ above SM ? **PASS**

Ug4 quiescent: negligible Ug4 contribution at $A_{\text{AGN}} = 1$? **PASS**

6. IceCube-Gen2 Detectability

Current IceCube sensitivity: $F_{3s} \sim 10^{10} \text{ TeV cm}^2 \text{ s}^{-1} \text{ sr}^{-1}$ for Sgr A* point source.

UQFF prediction: $\tau = 0.3\%$ excess above SM corona prediction.

Not detectable by IceCube-Gen2 (~10-year run). However, an exaggerated AGN-active Sgr A* state ($A_{\text{AGN}} \gg 10$) would push Ug4 enhancement to ~5% potentially within 3s of Gen2.

Summary

Prediction	UQFF Value	Standard Model	UQFF Signature
TRZ enhancement	+1.0%	0%	UQFF-specific
Flavor ratio	1:1:1 (1.001)	1:1:1	Unmeasurable
Quiescent Ug4	negligible	None	Consistent
AGN-active Ug4	+0.3-5%	0%	Conditional
Phase 3 validation	4/4 PASS	N/A	Verified

Source: MAIN1CoAnQi.cpp Batch 21 (NeutrinoSEModule) | validatephase3.py | fTRZ = 0.01 | IceCube constraints.Groups[1].Value "PAPER_0:D3" -f \$n

Abstract

The neutrino spectral energy distribution (SED) from a black hole system particularly Sagittarius A* probes high-energy processes in the UQFF framework through three channels: Hawking neutrino emission ($TUQFF = 0.99 TH$), AGN corona pp/p interactions (Ug4-mediated), and UQFF Toroidal Resonance Zone (TRZ) emission. Batch 21 of MAIN1CoAnQi.cpp implements the NeutrinoSEModule returning flux predictions at $E = 1 TeV-10 PeV$ for comparison with IceCube-Gen2 sensitivity. The UQFF predicts a neutrino flux excess of 0.3% above the standard model corona prediction, driven by $fTRZ = 0.01$.

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For stellar and SMBH systems, **Hawking neutrinos are negligible** ($TH \sim 10^4 K$ $E \sim kB TH \ll 1 eV$). The interesting band is TeV-PeV cosmic neutrinos from AGN corona interactions.

2. The UQFF Neutrino SED

2.1 Standard Model Corona Prediction

For Sgr A* in active state with corona luminosity $L_{corona} = 10^4 L_{Edd}$:

$$\Phi_{\nu}^{\text{SM}}(E_{\nu}) = \Phi_0 \left(\frac{E_{\nu}}{\text{TeV}} \right)^{-\gamma} e^{-E_{\nu}/E_{\text{cut}}}$$

With $\gamma = 2.2$, $E_{\text{cut}} = 5 \text{ PeV}$.

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The Toroidal Resonance Zone factor $f_{\text{TRZ}} = 0.01$ enhances corona pair production:

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The Ug4 energy density ($3.352941 \times 10^{10} \text{ J/m}^3$ baseline) additionally contributes to pion production:

$$\Delta \Phi_{\nu}^{\text{Ug4}}(E_{\nu}) = f_{\text{Ug4}} \cdot \Phi_{\nu}^{\text{SM}} \cdot \frac{U_{\text{g4}}}{U_{\text{g4,ref}}}$$

For current Sgr A* quiescent state ($A_{\text{AGN}} = 1$, $e^{-\tau}$ near-unity): $f_{\text{Ug4}} \ll f_{\text{TRZ}}$.

3. Multi-Flavor Ratio

Standard model predictions: $\Phi_e : \Phi_{\mu} : \Phi_{\tau} = 1:2:0$ at production $\Phi_e : \Phi_{\mu} : \Phi_{\tau} = 1:1:1$ after oscillation.

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$$(\Phi_e : \Phi_{\mu} : \Phi_{\tau})_{\text{UQFF}} = (1 : 1 : 1) \times (1 + 0.001 \times f_{\text{TRZ}})$$

The 26D mixing is degenerate in flavor at the detector level **no measurable deviation from 1:1:1** at IceCube sensitivity. This prediction is consistent with IceCube observations showing approximate (but not exact) flavor democracy.

4. NeutrinoSEModule (Batch 21)

From MAIN/CoAnQi.cpp Batch 21:

```
// namespace SOURCE_BATCH21
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validate_phase3.py performs phase-3 cross-validation of Batch 21 outputs against uqff_validation_test.py:

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Multi-flavor test: (1:1:1) ratio ? **PASS**

TRZ enhancement: ? = 1.0% above SM ? **PASS**

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Current IceCube sensitivity: $F_{3s} \lesssim 10^{-10} \text{ TeV cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$ for Sgr A* point source.

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Not detectable by IceCube-Gen2 (~10-year run). However, an exaggerated AGN-active Sgr A* state ($A_{\text{AGN}} \gg 10$) would push Ug4 enhancement to ~5% potentially within 3s of Gen2.

Summary

Prediction	UQFF Value	Standard Model	UQFF Signature
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Quiescent Ug4	negligible	None	Consistent
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Phase 3 validation	4/4 PASS	N/A	Verified

Source: MAIN1CoAnQi.cpp Batch 21 (NeutrinoSEModule) | validatephase3.py | $f_{\text{TRZ}} = 0.01$ | IceCube constraints.Groups[1].Value

Abstract

The neutrino spectral energy distribution (SED) from a black hole system particularly Sagittarius A* probes high-energy processes in the UQFF framework through three channels: Hawking neutrino emission ($T_{\text{UQFF}} = 0.99 \text{ TH}$), AGN corona pp/p interactions (Ug4-mediated), and UQFF Toroidal Resonance Zone (TRZ) emission. Batch 21 of MAIN1CoAnQi.cpp implements the NeutrinoSEModule returning flux predictions at $E = 1 \text{ TeV} - 10 \text{ PeV}$ for comparison with IceCube-Gen2 sensitivity. The UQFF predicts a neutrino flux excess of 0.3% above the standard model corona prediction, driven by $f_{\text{TRZ}} = 0.01$.

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Source: MAIN1CoAnQi.cpp Batch 21 (NeutrinoSEDModule) | validatephase3.py | fTRZ = 0.01 | IceCube constraints .Groups[1].Value "PAPER_0:D3" -f \$n

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Source: MAIN1CoAnQi.cpp Batch 21 (NeutrinoSEDModule) | validatephase3.py | $f_{TRZ} = 0.01$ | IceCube constraints

UQFF computed: UQFF-modified neutrino mass shift $\delta m_{\nu} = 2.85e-4 \sqrt{v/MN}$; for $MN = 1.0e14$ GeV: $\delta m_{\nu} = 5.1e-16$ GeV.